

Vibe Audio Protocol (V.A.P.)

Version: 3.1 (Draft Standard) **Date:** October 28, 2026 **Status:** Reference Specification

1. Introduction

The Vibe Audio Protocol (V.A.P.) is a universal metadata standard designed to capture the holographic identity of audio assets. Unlike legacy standards (ID3) which focus on bibliographic data, V.A.P. focuses on **Experiential Data**—quantifying the structural, emotional, and environmental properties of sound.

This document defines the **9-Pillar Taxonomy** and the **Logic Architecture** required to implement V.A.P. in software systems.

2. Logic Architecture (The Scoring Engine)

The generation of V.A.P. metadata occurs in three distinct phases:

Phase I: Physical Analysis (DSP Layer)

Objective: Extract objective physical properties from the audio signal.

1. Structural Logic

- **Kick Transient Profile (ms):** Measured from signal onset (-60dB) to peak amplitude.
 - < 10ms: **Sharp (Click)** (Tech/Metal)
 - 10-30ms: **Punch (Thud)** (Pop/Rock)
 - > 30ms: **Boom (Sub)** (Trap/808)
- **Syncopation Index (0.0-1.0):** Ratio of transient energy on weak beats vs. strong beats.
 - 0.0: Marching/Motorik
 - 1.0: Polyrhythmic/Jazz

2. Tonal Logic

- **Dissonance Density (%):** Percentage of total duration containing high-tension intervals (Minor 2nd, Tritone).
 - < 10%: **Consonant**
 - > 40%: **Dissonant**

3. Timbral Logic

- **Spectral Centroid (Hz):** Center of gravity of the frequency spectrum.
 - < 200Hz: **Dark/Muddy**
 - 200-2000Hz: **Warm/Body**
 - > 2000Hz: **Bright/Airy**

Phase II: Psychological Analysis (NLP & ML Layer)

Objective: Infer subjective human experience using Machine Learning models.

4. Linguistic Logic

- **Explicit Severity Tier:**
 - **Tier 1 (Clean):** 0 Safety List matches.
 - **Tier 4 (Severe):** Hate speech or extreme violence markers.

5. Affective Logic (Thayer Model)

- **Valence (-1.0 to +1.0):** Derived from Key (Major/Minor) + Sentiment Analysis.
- **Arousal (0.0 to 1.0):** Derived from RMS Amplitude + BPM.
 - **> 0.8: High Arousal** (Rage/Euphoria)
 - **< 0.2: Low Arousal** (Sleep/Calm)

6. Contextual Logic (Bayesian Probability)

- **Scenario Confidence:** $P(\text{Scenario} \mid \text{Audio_Features})$.
 - Example: If Genre=Synthwave AND Key=Minor AND BPM=100, THEN Scenario=Night_Drive (Confidence > 85%).

Phase III: Environmental & Social Analysis (I/O Layer)

Objective: Format data for hardware output and community validation.

7. Photometric Logic (Lighting)

- **Chromatic Mapping:** Frequency to Wavelength conversion.
 - Low Freq (Bass) -> Long Wavelength (Red/Infrared)
 - High Freq (Treble) -> Short Wavelength (Blue/UV)

8. Kinetic Logic (Biometrics)

- **MET Score (Metabolic Equivalent):**
 - BPM < 60: **1.0 MET** (Rest)
 - BPM > 140: **8.0+ MET** (Sprint/HIIT)

9. Genealogical Logic (Tribe)

- **Authenticity Ratio:** (Positive Votes by Tribe Members / Total Votes).
 - **> 80%: True/Authentic**
 - **< 40%: Poser/Mainstream**

3. Implementation Guidelines

3.1 JSON Schema

All V.A.P. data must be encapsulated in a vap_object JSON structure embedded within the file header (ID3v2 TXXX frame or Vorbis Comment).

3.2 Backward Compatibility

Systems that do not support V.A.P. will simply ignore the metadata header and play the audio file as standard.

3.3 Hardware Certification

Lighting fixtures and biometric devices (e.g., AuraOrb) must be certified to read **Pillar 7 (Photometric)** and **Pillar 8 (Kinetic)** data streams respectively.